

① $(f+g)(x) = f(x) + g(x)$

$a \cdot f(x) = a \cdot f(x)$

+) 0 polynomial is in V since $0(2) = 0$

++) $f(2) = 0$ $g(2) = 0 \Rightarrow f(2) + g(2) = 0 \Rightarrow f+g \in V$

+++) Let $f \in V$ and $a \in \mathbb{R}$

$f(2) = 0$

$a \cdot f(2) = a \cdot 0 = 0$

$a \cdot f \in V$

V is a subspace of V or V is a vector space with respect to the standard operations in P_3

$$(2) \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 3 \\ -2 & -6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a-2b & 3a-6b \\ c-2d & 3c-6d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \begin{array}{l} a-2b=0 \\ 3a-6b=0 \\ c-2d=0 \\ 3c-6d=0 \end{array} \quad \begin{array}{l} a=2b \\ c=2d \end{array}$$

$\Rightarrow$ subspace

$$y \quad b=d=0 \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in M_{22}$$

$$\begin{bmatrix} 2b & b \\ 2d & d \end{bmatrix}, \begin{bmatrix} 2x & x \\ 2y & y \end{bmatrix} \in M_{22}$$

$$\begin{bmatrix} 2b & b \\ 2d & d \end{bmatrix} + \begin{bmatrix} 2x & x \\ 2y & y \end{bmatrix} = \begin{bmatrix} 2(b+x) & b+x \\ 2(d+y) & d+y \end{bmatrix} \in M_{22}$$

$$\Rightarrow \begin{bmatrix} 2b & b \\ 2d & d \end{bmatrix} \in M_{22} \quad x \in \mathbb{R}$$

$$x \begin{bmatrix} 2b & b \\ 2d & d \end{bmatrix} = \begin{bmatrix} 2bx & bx \\ 2dx & dx \end{bmatrix} \in M_{22}$$

A is subspace of M_{22}

(3)

$$c_1, c_2, c_3 \in \mathbb{R}$$

$$c_1(1+x^2) + c_2(x+x^3) + c_3(3-2x+3x^2+12x^3) = 0$$

$$\cancel{c_1} + \cancel{c_1}x^2 + \cancel{c_2}x + c_2x^3 + \cancel{3c_3} - \cancel{2x}c_3 + \cancel{3x^2}c_3 + 12x^3c_3 = 0$$

$$(c_1 + 3c_3) + (c_2 - 2c_3)x + (c_1 + 3c_3)x^2 + (c_2 + 12c_3)x^3 = 0 \quad) = 0$$

$$c_1 + 3c_3 = 0$$

$$c_2 = 2c_3$$

$$c_2 - 2c_3 = 0$$

$$2c_3 + 12c_3 = 0$$

$$c_1 + 3c_3 = 0$$

$$14c_3 = 0$$

$$c_2 + 12c_3 = 0$$

$$c_2 = 0$$

$$c_1 = 0$$

S is lin. indep.

$$P(x) = (a+3c) + (b-2c)x + (a+3c)x^2 + (b+12c)x^3 \in W$$

$$a, b, c \in \mathbb{R} \Rightarrow a+3c, b-2c, a+3c, b+12c \in \mathbb{R}$$

$$W \subseteq \text{span}(S)$$

$$\text{span}(S) \subseteq W$$

$$W = \text{span}(S)$$

$$c_1 \begin{bmatrix} 1 & 2 \\ -1 & 1 \\ 3 & 0 \end{bmatrix} + c_2 \begin{bmatrix} 4 & 2 \\ -6 & 1 \\ 0 & 1 \end{bmatrix} + c_3 \begin{bmatrix} 0 & 1 \\ 1 & -1 \\ 2 & 2 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 7 \\ 5 & 2 \\ -1 & 6 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$c_1 + 4c_2 = 0$$

$$2c_1 + 2c_2 + c_3 + 7c_4 = 0$$

$$-c_1 - 6c_2 + c_3 + 5c_4 = 0$$

$$c_1 + c_2 - c_3 + 2c_4 = 0$$

$$3c_1 + 2c_3 - c_4 = 0$$

$$c_2 + 2c_3 + 6c_4 = 0$$

$$\begin{bmatrix} c_1 & c_2 & c_3 & c_4 \\ 1 & 4 & 0 & 0 \\ -2 & 2 & 1 & 7 \\ -1 & -6 & 1 & 5 \\ 1 & 1 & -1 & 2 \\ 3 & 0 & 2 & -1 \\ 0 & 1 & 2 & 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2R_1 + R_2 \rightarrow R_2$$

$$R_1 + R_2 \rightarrow R_3$$

$$-R_1 + R_4 \rightarrow R_4$$

$$-R_1 + R_5 \rightarrow R_5$$

$$\begin{bmatrix} 1 & 4 & 0 & 0 \\ 0 & -6 & 1 & 7 \\ 0 & -2 & 1 & 5 \\ 0 & -3 & -1 & 2 \\ 0 & -12 & 2 & -1 \\ 0 & 1 & 2 & 6 \end{bmatrix}$$

$$-\frac{R_2}{3} + R_3 \rightarrow R_3$$

$$-\frac{R_2}{2} + R_4 \rightarrow R_4$$

$$-2R_2 + R_5 \rightarrow R_5$$

$$\frac{R_2}{6} + R_6 \rightarrow R_6$$

$$\begin{bmatrix} 1 & 4 & 0 & 0 \\ 0 & -6 & 1 & 7 \\ 0 & 0 & 2/3 & 8/3 \\ 0 & 0 & -3/2 & -3/2 \\ 0 & 0 & 0 & -15 \\ 0 & 0 & 13/6 & 43/6 \end{bmatrix}$$

$$\frac{9R_3}{4} + R_4 \rightarrow R_4$$

$$-\frac{13R_3}{4} + R_6 \rightarrow R_6$$

$$\begin{bmatrix} 1 & 4 & 0 & 0 \\ 0 & -6 & 1 & 7 \\ 0 & 0 & 2/3 & 8/3 \\ 0 & 0 & 0 & 9/2 \\ 0 & 0 & 0 & -15 \\ 0 & 0 & 0 & -9/6 \end{bmatrix}$$

$$-\frac{10R_4}{3} + R_5 \rightarrow R_5$$

$$\rightarrow$$

$$-\frac{R_3}{3} + R_6 \rightarrow R_6$$

$$\begin{bmatrix} 1 & 4 & 0 & 0 \\ 0 & -6 & 1 & 7 \\ 0 & 0 & 2/3 & 8/3 \\ 0 & 0 & 0 & 9/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\frac{9c_4}{2} = 0 \Rightarrow c_4 = 0$$

$$\frac{2c_3}{3} + \frac{8c_4}{3} = 0 \Rightarrow c_3 = 0$$

$$-6c_2 + c_3 + 7c_4 = 0 \Rightarrow c_2 = 0$$

$$c_1 + 4c_2 = 0 \Rightarrow c_1 = 0$$

linearly independent

$$\textcircled{5} \left[\begin{array}{ccccc} 1 & 3 & 2 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 4 & 1 & 3 \\ 0 & 3 & 3 & 1 & 2 \\ 1 & 1 & 5 & 1 & 4 \end{array} \right] \begin{array}{l} -R_1 + R_3 \rightarrow R_3 \\ \xrightarrow{\quad} \\ -R_1 + R_5 \rightarrow R_5 \end{array}$$

$$\left[\begin{array}{ccccc} 1 & 3 & 2 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & -2 & 2 & 0 & 2 \\ 0 & 3 & 3 & 1 & 2 \\ 0 & -2 & 3 & 0 & 3 \end{array} \right] \begin{array}{l} -R_3 + R_5 \rightarrow R_5 \\ \xrightarrow{\quad} \\ \frac{3R_3}{2} + R_4 \rightarrow R_4 \end{array}$$

$$\left[\begin{array}{ccccc} 1 & 3 & 2 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & -2 & 2 & 0 & 2 \\ 0 & 0 & 6 & 1 & 6 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right] \begin{array}{l} -\frac{R_3}{2} \rightarrow R_3 \\ \xrightarrow{\quad} \\ R_2 \leftrightarrow R_3 \end{array}$$

$$\left[\begin{array}{ccccc} 1 & 3 & 2 & 1 & 1 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 6 & 1 & 6 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right] \begin{array}{l} -6R_3 + R_4 \rightarrow R_4 \\ \xrightarrow{\quad} \\ -R_3 + R_5 \rightarrow R_5 \end{array}$$

$$\left[\begin{array}{ccccc} 1 & 3 & 2 & 1 & 1 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -5 & 5 \\ 0 & 0 & 0 & -1 & 1 \end{array} \right] \xrightarrow{-\frac{R_4}{5} \rightarrow R_4}$$

$$\left[\begin{array}{ccccc} 1 & 3 & 2 & 1 & 1 \\ 0 & 1 & -1 & 0 & -1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{array} \right]$$

$$R_4 + R_5 \rightarrow R_5$$

$$\left[\begin{array}{ccccc} \textcircled{1} & 3 & 2 & 1 & 1 \\ 0 & \textcircled{1} & -1 & 0 & -1 \\ 0 & 0 & \textcircled{1} & 1 & 0 \\ 0 & 0 & 0 & \textcircled{1} & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

is not a base

4 pivot $\neq$ 5 dimension

base = 4)

U doesn't span $\mathbb{R}^5$

(6)

$$|\lambda I - A| = \left| \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix} \right| = \begin{vmatrix} \lambda-3 & 0 & 0 \\ 0 & \lambda-2 & -1 \\ +1 & 0 & \lambda-2 \end{vmatrix}$$

$$= (\lambda-3) \begin{vmatrix} \lambda-2 & -1 \\ 0 & \lambda-2 \end{vmatrix} \Rightarrow (\lambda-3)(\lambda^2 - 4\lambda + 4)$$

$$\Rightarrow (\lambda-3)(\lambda-2)^2$$

$$\lambda_1 = 3 \quad \lambda_2 = 2 \quad \lambda_3 = 2 \Rightarrow \text{eigen values}$$

$$(\lambda I - A)x = 0 \quad \lambda = 3$$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right] \xrightarrow{R_3 \leftrightarrow R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$x_1 + x_3 = 0$$

$$x_3 = a$$

$$x_1 = -a$$

$$x_2 - x_3 = 0$$

$$x_2 = a$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = a \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

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⑥ $(2I - A)x = 0$ $\lambda = 2$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_1 + R_3 \rightarrow R_3$
 $\xrightarrow{-R_1 \Rightarrow R_1}$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$x_1 = 0$
 $x_2 = a$
 $x_3 = 0$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = a \underline{\underline{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}}$$

(7)

$$A = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 3 & 2 \\ -2 & 7 & 6 \end{bmatrix}$$

$$\begin{aligned} a/ \quad A_{11} &= (-1)^{1+1} (3 \cdot 6 - 2 \cdot 7) = 4 \\ A_{12} &= (-1)^{1+2} (0 \cdot 6 - (-2 \cdot 2)) = -4 \\ A_{13} &= (-1)^{1+3} (0 \cdot 7 - (-2 \cdot 3)) = 6 \\ A_{21} &= (-1)^{2+1} (0 \cdot 6 - (-4 \cdot 7)) = 28 \\ A_{22} &= (-1)^{2+2} (1 \cdot 6 - (-2 \cdot (-4))) = -2 \\ A_{23} &= (-1)^{2+3} (1 \cdot 7 - (-2 \cdot 0)) = -7 \\ A_{31} &= (-1)^{3+1} (0 \cdot 2 - (-4 \cdot 3)) = 12 \\ A_{32} &= (-1)^{3+2} (1 \cdot 2 - (-4 \cdot 0)) = -2 \\ A_{33} &= (-1)^{3+3} (1 \cdot 3 - 0 \cdot 0) = 3 \end{aligned}$$

$$\text{cof}(A) = \begin{bmatrix} 4 & -4 & 6 \\ 28 & -2 & -7 \\ 12 & -2 & 3 \end{bmatrix}$$

$$\text{adj}(A) = \begin{bmatrix} 4 & 28 & 12 \\ -4 & -2 & -2 \\ 6 & -7 & 3 \end{bmatrix}$$

$$\begin{aligned} b/ \quad \begin{vmatrix} 1 & 0 & -4 \\ 0 & 3 & 2 \\ -2 & 7 & 6 \end{vmatrix} &\Rightarrow \det A (18 + 0 + 0) - (24 + 14 + 0) \\ &= 18 - 38 \\ &= -20 \end{aligned}$$

$$c/ \quad A^{-1} = \frac{1}{\det A} \cdot \text{adj}(A)$$

$$\frac{1}{-20} \begin{bmatrix} 4 & 28 & 12 \\ -4 & -2 & -2 \\ 6 & -7 & 3 \end{bmatrix} = \begin{bmatrix} -1/5 & -7/5 & -3/5 \\ 1/5 & 1/10 & 1/10 \\ -3/10 & 7/20 & -3/20 \end{bmatrix}$$

$$\textcircled{2} \quad \begin{bmatrix} -5 & 2 & -2 & 1 \\ 2 & -1 & 2 & -2 \\ 5 & -2 & 3 & -1 \\ -6 & 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -10 \\ -9 \\ 7 \\ -14 \end{bmatrix}$$

$$x_1 \Rightarrow \begin{vmatrix} -10 & 2 & -2 & 1 \\ -9 & -1 & 2 & -2 \\ 7 & -2 & 3 & -1 \\ -14 & 2 & -2 & 1 \end{vmatrix} \Rightarrow \frac{1 \Delta x_1}{|A|} = \frac{12}{3} = 4 \quad x_1 = 4$$

$$\begin{vmatrix} -5 & 2 & -2 & 1 \\ 2 & -1 & 2 & -2 \\ 5 & -2 & 3 & -1 \\ -6 & 2 & -2 & 1 \end{vmatrix}$$

$$\begin{array}{l} R_4 + R_3 \rightarrow R_3 \\ 2R_4 + R_2 \rightarrow R_2 \\ -R_4 + R_1 \rightarrow R_1 \end{array} \quad \begin{bmatrix} 4 & 0 & 0 & 0 \\ -37 & 3 & -2 & 0 \\ -7 & 0 & 1 & 0 \\ -14 & 2 & -2 & 1 \end{bmatrix} \xrightarrow{R_3 + R_2 \rightarrow R_2} \begin{bmatrix} 4 & 0 & 0 & 0 \\ -51 & 3 & 0 & 0 \\ -7 & 0 & 1 & 0 \\ -14 & 2 & -2 & 1 \end{bmatrix}$$

$$\det = 4 \cdot 3 \cdot 1 \cdot 1 = \underline{12}$$

$$\begin{bmatrix} -5 & 2 & -2 & 1 \\ 2 & -1 & 2 & -2 \\ 5 & -2 & 3 & -1 \\ -6 & 2 & -2 & 1 \end{bmatrix} \xrightarrow{\begin{array}{l} R_4 + R_3 \rightarrow R_3 \\ 2R_4 + R_2 \rightarrow R_2 \\ -R_4 + R_1 \rightarrow R_1 \end{array}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -16 & 3 & -2 & 0 \\ -1 & 0 & 1 & 0 \\ -6 & 2 & -2 & 1 \end{bmatrix} \xrightarrow{2R_3 + R_2 \rightarrow R_2}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ -18 & 3 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -6 & 2 & -2 & 1 \end{bmatrix}$$

$$\det A = 1 \cdot 3 \cdot 1 \cdot 1 = 3$$

①

(3)

$$x_2 \Rightarrow \begin{vmatrix} -5 & -10 & -2 & 1 \\ 2 & -9 & 2 & -2 \\ 5 & 2 & 3 & -1 \\ -6 & -14 & -2 & 1 \end{vmatrix} \Rightarrow \frac{|\Delta x_2|}{|A|} = \frac{-3}{3} = -1$$

$$\det A = 3$$

$$\begin{bmatrix} -5 & -10 & -2 & 1 \\ 2 & -9 & 2 & -2 \\ 5 & 2 & 3 & -1 \\ -6 & -14 & -2 & 1 \end{bmatrix} \begin{array}{l} R_4 + R_3 \rightarrow R_3 \\ 2R_4 + R_2 \rightarrow R_2 \\ -R_4 + R_1 \rightarrow R_1 \end{array}$$

$$\begin{bmatrix} 1 & 4 & 0 & 0 \\ -10 & -37 & -2 & 0 \\ -1 & -7 & 1 & 0 \\ -6 & -14 & -2 & 1 \end{bmatrix}$$

$$2R_3 + R_2 \rightarrow R_2 \begin{bmatrix} 1 & 4 & 0 & 0 \\ -12 & -51 & 0 & 0 \\ -1 & -7 & 1 & 0 \\ -6 & -14 & -2 & 1 \end{bmatrix}$$

$$\frac{4}{51} R_2 + R_1 \rightarrow R_1 \begin{bmatrix} \frac{3}{51} & 0 & 0 & 0 \\ -12 & -51 & 0 & 0 \\ -1 & -7 & 1 & 0 \\ -6 & -14 & -2 & 1 \end{bmatrix}$$

$$\det = \frac{3}{51} \cdot (-51) \cdot 1 \cdot 1 = -3$$

$$x_3 \Rightarrow \begin{vmatrix} -5 & 2 & -10 & 1 \\ 2 & -1 & -9 & -2 \\ 5 & -2 & 2 & -1 \\ -6 & 2 & -14 & 1 \end{vmatrix} \rightarrow \frac{|\Delta x_3|}{|A|} = \frac{-9}{3} = -3$$

$$\det A = 3$$

$$\begin{array}{l} R_4 + R_3 \rightarrow R_3 \\ 2R_4 + R_2 \rightarrow R_2 \\ -R_4 + R_1 \rightarrow R_1 \end{array} \begin{bmatrix} 1 & 0 & 4 & 0 \\ -10 & 3 & -37 & 0 \\ -1 & 0 & -7 & 0 \\ -6 & 2 & -14 & 1 \end{bmatrix} \begin{array}{l} \frac{-37}{7} R_3 + R_2 \rightarrow R_2 \\ \frac{4}{7} R_3 + R_1 \rightarrow R_1 \end{array}$$

$$\begin{bmatrix} \frac{3}{7} & 0 & 0 & 0 \\ -\frac{43}{7} & 3 & 0 & 0 \\ -1 & 0 & -7 & 0 \\ -6 & 2 & -14 & 1 \end{bmatrix}$$

$$\det = \frac{3}{7} \cdot 3 \cdot (-7) \cdot 1 = -9$$

(2)

(8)

$$x_4 \Rightarrow \begin{vmatrix} -5 & 2 & -2 & -10 \\ 2 & -1 & 2 & -9 \\ 5 & -2 & 3 & 7 \\ -6 & 2 & -2 & -14 \end{vmatrix} = \frac{|\Delta x_4|}{|A|} = \frac{18}{3} = 6$$

$$\det A = 3$$

$$(x_1, x_2, x_3, x_4) = (4, -1, -3, 6)$$

(9)

a $\rightarrow$ True

b $\rightarrow$ True

c $\rightarrow$ False

d $\rightarrow$ False

e $\rightarrow$ True